THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE UNIT TEST SET 2

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The vessels that carry water and minerals from roots to leaves are the: 1

(a) veins of animals **(b)** xylem **(c)** stomata **(d)** phloem

2. The size of a shadow becomes larger when the object is moved: 1

(a) closer to the screen **(b)** farther from the source

(c) behind the screen **(d)** closer to the light source

3. The shortest day in the northern hemisphere is on about: 1

(a) 21 March **(b)** 21 June **(c)** 23 September **(d)** 22 December

4. Rhizobium bacteria help the plant by: 1

(a) giving light **(b)** making sugar

(c) fixing atmospheric nitrogen **(d)** giving water

5. A surface that reflects light well is: 1

(a) a rough stone **(b)** a mirror **(c)** soil **(d)** a black cloth

6. The cycle of Moon's phases repeats in about: 1

(a) 7 days **(b)** 365 days **(c)** 1 day **(d)** 29.5 days

7. Animals get food directly or indirectly from: 1

(a) minerals **(b)** soil only **(c)** plants **(d)** air

8. The size of the image in a plane mirror is: 1

(a) same as the object **(b)** smaller

(c) larger **(d)** double

9. A solar eclipse occurs on a: 1

(a) any day **(b)** quarter moon **(c)** full moon day **(d)** new moon day

10. The green pigment in leaves is: 1

(a) insulin **(b)** carotene only **(c)** haemoglobin **(d)** chlorophyll

11. Which of the following is a luminous object? 1

(a) the Sun **(b)** a book **(c)** a mirror **(d)** the Moon

12. One revolution of the Earth around the Sun takes about: 1

(a) 30 days **(b)** 365 and a quarter days
(c) 7 days **(d)** 24 hours

13. The vessels that carry food from the leaves to other parts are the: 1

(a) phloem **(b)** root hairs **(c)** xylem **(d)** stomata

14. The darkest part of a shadow is the: 1

(a) penumbra **(b)** image **(c)** halo **(d)** umbra

15. Days and nights are equal on about: 1

(a) 15 August **(b)** 1 January
(c) 21 June and 22 December **(d)** 21 March and 23 September

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Write the date of two solstices. 2

17. What is symbiosis? Give an example. 2

18. What is a periscope used for? 2

19. When does a solar eclipse occur? 2

20. Name the raw materials needed for photosynthesis. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain the formation of umbra and penumbra. 3

22. Why are eclipses not seen every month? 3

23. Describe the activity to show that sunlight is needed for photosynthesis. 3

24. Describe how a shadow changes in size when an object is moved towards the screen. 3

OR

Describe why the Earth has seasons. 3

25. Explain symbiotic nutrition with the example of lichen. 3

OR

Explain how a periscope works. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe a kaleidoscope: how it is made and the pattern formed. 5

OR

Describe the features of the Moon and compare with the Earth. 5

27. Describe the experiments to show that chlorophyll, light and carbon dioxide are necessary for photosynthesis. 5

OR

Describe the working of a pinhole camera. Explain why the image is inverted. 5

END OF QUESTION PAPER — Total: 50 Marks

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